Overhearing Agents

Dimensions of Overhearing

- First: survey NLP + HCI papers to create a grounded taxonomy of overhearing

	Dimension	Value	Example(s)
User Interaction	Initiative	Always Active	Always-listening voice assistants (Tabassum et al. 2019; Zargham et al. 2022)
		User-Initiated	Wordcraft (Yuan et al. 2022)
		Post-Hoc Analysis	More to Meetings (McGregor and Tang 2017)
		Rule-Based	Apple Intelligence (Apple 2025)
	Input Modality	Audio	Bardo (Padovani, Ferreira, and Lelis 2017), LlamaPIE (Chen et al. 2025)
		Text	Code "Copilots" (Barke, James, and Polikarpova 2023)
		Video	EgoExo4D (Grauman et al. 2024)
	Interfaces	Web and Desktop	Proactive Search (Andolina et al. 2018), ProactiveAgent (Lu et al. 2024)
		Wearable Devices	LlamaPIE (Chen et al. 2025), Meta smart glasses (Meta 2024)
		Smart Home	Home voice assistants (Tabassum et al. 2019; Zargham et al. 2022; Hwang et al. 2023)
System Design	State	Read-Only	Proactive Search (Andolina et al. 2018)
		Read-Write	More to Meetings (McGregor and Tang 2017)
	Timeliness	Real-Time	Bardo (Padovani, Ferreira, and Lelis 2017)
		Asynchronous	ReDel (Zhu, Dugan, and Callison-Burch 2024)
	Interactivity	Foreground	Most overhearing agents
		Background	AI agent memory (Park et al. 2023), DocPrompting (Zhou et al. 2023)

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Challenge Domains

- Let's learn more about overhearing agents by making some!
- 3 challenge domains: D&D, clinical agenda-setting, and cooking

Domain	Description	Challenges
Dungeons & Dragons	Assist a Dungeon Master with game tasks like game entity lookup or selecting music	Intervention-point prediction, many-speaker environments
Clinical Agenda-Setting	Identify and manage a clinical agenda based on patient-provider conversations	On-device & HIPAA-compliant processing restrictions
Cooking	Offer timely suggestions to a user who's cooking a recipe	Streaming video input